

## 1. One-line summary

A field-first web app that discovers real-world problems in a user's chosen domain, ranks them by measurable pain, generates layered AI solutions, creates phased execution roadmaps with time estimates, and auto-produces developer-ready pre-build documents on demand.

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## 2. Target users

- Founders and startup teams looking for validated problem-solution fits.
  - Product managers and designers needing prioritized problem lists + roadmaps.
  - Engineers seeking actionable specs and architecture scaffolding.
  - Community-builders, researchers, and domain specialists (education, healthcare, finance, etc.).
  - Power users who want to log, implement, or claim solved problems.
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## 3. Core value proposition

Turn domain signals into prioritized, actionable projects (solution + plan + docs) in minutes — not ideas or essays, but execution-ready artifacts (roadmaps, PRDs, tech docs, design stubs).

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## 4. Field-first onboarding (must)

- On first-run user selects one or more domains (Education, Finance, Business, Tech, Healthcare, Community/Social, Custom).
- Selections persist in profile and bias: discovery sources, language / tone, metrics emphasized, suggested stacks, and roadmap granularity.
- UX: simple multi-select with examples and one “refine later” toggle.

**Effect:** every discovery, ranking, and roadmap is context-aware.

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## 5. Problem discovery (data + ranking)

### Sources (multi-source aggregator)

- Public forums & technical feeds (e.g., Reddit, Stack Overflow, Hacker News, GitHub Issues, Product Hunt comments), curated newsletters & blogs, social thread harvest (Twitter/X threads), and specialized APIs (e.g., Perplexity).
- Crawlers + API connectors with configurable source weights per domain.

### Extraction

- NLP pipeline to extract problem statements, timestamps, discussion volume, sentiment, and solution attempts (if any).

- Normalize duplicates, cluster similar problems, aggregate historical context.

### Pain Index (ranking)

- Compute per-cluster Pain Index (Pi). Use the provided structure:

$$Pi = (\text{Volume\_of\_Mentions} \times \text{Avg\_Sentiment\_Complexity}) / (\text{Age\_in\_Hours}) \times \text{Source\_Authority}$$

- Implementation details:
  - **Volume\_of\_Mentions:** unique mentions across sources in a lookback window (configurable: 7/30/90 days).
  - **Avg\_Sentiment\_Complexity:** composite score combining sentiment (negativity weight) and semantic complexity (how technical/ambiguous the problem is).
  - **Age\_in\_Hours:** time since first mention in cluster; lower age boosts Pi.
  - **Source\_Authority:** source authority multiplier from a domain-specific authority map.

### Output in UI (problem card)

- Short problem summary (1–2 lines)
- Pain Index (numeric + bar)
- Difficulty (estimate: easy/medium/hard)
- Field tag(s)
- Trending score (delta of Pi over last N days)
- Quick metadata (top sources, first-seen date)

### Feed behavior

- Default sorted by Pain Index; filters for Domain, Difficulty, Source, Trending; search and save/bookmark.

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## 6. AI solution generation (layered, actionable)

For any selected problem, the app generates in layers:

1. **One-line solution headline** — concise value prop.
2. **High-level approach** — 3–5 bullet conceptual plan.
3. **Execution approach** — prioritized, phased tasks (precedes roadmap button).
4. **Recommended tech stack** — Frontend, Backend, Database, Infra, DevOps, 3rd-party services (Auth, Payments, ML tools, observability). Provide options (primary + 2 alternatives).
5. **Estimated difficulty & risk** — engineering complexity, regulatory flags, data needs.
6. **Milestone test / validation experiments** — 2–3 low-cost experiments to validate assumptions.

Each layer is editable and exportable.

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## 7. Roadmap engine (critical)

**Trigger:** user clicks **Create Roadmap**.

### Roadmap output (phased)

- Auto-generated phases with:
  - Phase name
  - Objectives (measurable)
  - Tasks (ownerable, atomic)
  - Expected timespan (days/weeks)
  - Dependencies
  - Output artifacts (files, endpoints, tests)
  - Success criteria per phase

### Default phase template (example)

1. **Discovery & Validation** — 1–2 weeks
  - Tasks: stakeholder interviews, landing experiment, 5-user usability tests, quick metrics to validate demand.
  - Artifacts: validation notes, metric targets.
2. **Product Design** — 2–3 weeks
  - Tasks: wireframes, interactive prototype, user flows, accessibility checklist.
  - Artifacts: Figma file stub, UI spec.
3. **MVP Development** — 3–6 weeks
  - Tasks: API scaffolding, core features, front-end flow, basic infra.
  - Artifacts: deployable MVP, integration tests.
4. **Auth & Payments** — 1–2 weeks (if monetization needed)
  - Tasks: implement OAuth, subscriptions, billing flows, tax/compliance checks.
  - Artifacts: payment integration, test accounts.
5. **Testing & Edge Cases** — 1–2 weeks
  - Tasks: load tests, error handling, rate limiting, security review.
  - Artifacts: QA reports, runbook.
6. **Launch & Iteration** — ongoing
  - Tasks: analytics, feedback loop, feature prioritization.

- Artifacts: roadmap updates, KPIs.

### Time-span logic

- Roadmap engine uses difficulty, team-size input, and domain to estimate durations (can toggle for “solo”, “small team (2–5)”, “team (6+)”).

### Outputs

- Exportable roadmap (Markdown, CSV, PDF) and Kanban import (Trello/Github Projects/Notion templates).
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## 8. Document readiness system (pre-build checklist)

After a roadmap is generated UI shows “Documents you should have before building” with checkable items and descriptions:

- Product Requirements Document (PRD) — scope, acceptance criteria
- Design File — wireframes & high-fidelity mocks
- Technical Architecture Document — system diagrams & infra plan
- Auth Flow Document — flows, edge-cases, token policies
- Payment & Subscription Document — pricing tiers, billing logic
- Edge Case & Failure Handling Document — expected errors, mitigation
- Test Plan — QA & observability specifics

Each checklist item has a “Generate” button.

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## 9. Document generation (one-click)

Each document is produced by templates + LLM augmentation and must be:

- Structured (sections & subsections)
- Developer-ready (APIs, endpoints, data models)
- Editable and versioned
- Exportable: Markdown, PDF, and copy-to-clipboard

### Buttons & outputs

- **Generate PRD** — includes problem statement, goals, KPIs, user stories, acceptance criteria, out-of-scope.
- **Generate Design File (stub)** — Figma-ready component list, key screens, content map (downloadable JSON/Markdown that a designer can import).
- **Generate Tech & Architecture Doc** — diagrams (link to generated SVG/PNG), data models, API contract suggestions (OpenAPI stub), infra as code suggestions.

- **Generate Auth Flow** — sequence diagrams, token lifetimes, refresh policies, role matrix.
- **Generate Payment & Subscription File** — pricing matrix, billing flows, refund policies, tax considerations by region (high-level).
- **Generate Edge Case Handling File** — prioritized list of failures, recommended responses, observability hooks.

#### **Developer-ready details included**

- Example API routes, request/response shapes, DB schema stubs, example tests, CI/CD checklist, observability & monitoring endpoints.
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### **10. MVP scope (must-have vs nice-to-have)**

#### **Must-have (MVP)**

- Field-first onboarding & profile preferences
- Multi-source discovery connector (configurable seeds)
- Pain Index computation & ranked problem feed
- Layered AI solution for each problem
- Create Roadmap button generating phased roadmap (exportable)
- Document readiness checklist and at least PRD + Tech Doc generators (Markdown/PDF)
- Auth (email + OAuth) and user profile storage
- Basic progress tracking & “saved problems” list
- Export features (Markdown, PDF, Kanban CSV)

#### **Expansion (post-MVP)**

- Community “I solved this” flow + leaderboard + proof uploads
  - GitHub / CI integration (PR templates, repo scaffolds)
  - Notifications and scheduled discovery jobs (daily/weekly summaries)
  - Payments, premium tiers, team workspaces
  - Advanced analytics (A/B of validation experiments)
  - Native mobile apps (PWA + wrappers)
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### **11. Frontend & UX (human-first)**

- Mobile-first responsive SPA (Next.js + Tailwind recommended).
- Clear CTA funnel: Select Field → Problems Feed → Solution View → Create Roadmap → Generate Docs.

- Card-based problem feed with friction-minimizing microcopy.
  - Roadmap builder modal with single-click export and customizable timeline.
  - Document editor overlay (edit inline + version history).
  - Gamification (optional): points for “validated experiments”, badges for “roadmaps launched”, streaks for daily engagement.
  - Accessibility & localization-ready.
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## 12. Backend & AI architecture

### Components

- **Ingestion layer:** connectors + scrapers + queued parsers (rate-limited, source adapters).
- **Processing pipeline:** dedupe → clustering → metadata extraction → Pain Index calculation.
- **Index & storage:** Postgres / Supabase for relational data, vector DB (e.g., Pinecone/Weaviate) for embeddings & semantic search.
- **AI services:** LLMs for extraction, summarization, solution generation (replaceable: OpenAI/GPT, Anthropic/Claude, Blackbox). Use an orchestration layer to swap models and cache outputs.
- **Roadmap & doc generator:** rule-based templates + LLM filling; deterministic template engine with variables and post-edit hooks.
- **Async job worker:** for daily discovery, re-ranking, user reports (e.g., Celery / Cloud Tasks / Background workers).
- **Auth & billing:** OAuth + JWTs, Stripe (or regional equivalent) for subscriptions.
- **Observability:** logging, metrics (Prometheus), error tracking (Sentry), usage dashboards.

### Security & compliance

- Rate-limiting, API key protection for connectors, secure storage for PII, privacy policy for scraped content, GDPR-aware retention settings.
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## 13. Data model (high level)

- **User:** id, email, domains\_selected, role, subscription\_tier, saved\_problems, created\_roadmaps
- **ProblemCluster:** id, canonical\_text, pain\_index, difficulty, sources[], created\_at, last\_seen\_at, domain\_tags
- **Solution:** id, problem\_id, solution\_headline, layers (JSON), recommended\_stack (JSON)
- **Roadmap:** id, solution\_id, phases[], estimated\_time, team\_size\_assumption
- **Document:** id, type, content\_markdown, generated\_by, version, exports[]

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## 14. Auth & subscriptions

### Auth

- Email + password + magic link + OAuth (Google, GitHub).
- Role model: individual, team-admin, team-member.

### Subscription tiers

- **Free:** limited problem views/day, limited document generations, ads.
- **Pro:** unlimited generations, full roadmap exports, team workspace, priority model compute.
- **Enterprise:** custom connectors, SSO, on-premise options, SLA.

### Payment flows

- Integrate Stripe (or regional alternative), support trials, coupons, and prorations. Provide billing docs generator.

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## 15. Monetization strategy

- Freemium product-led growth: free for exploration, paid for execution artifacts + team features.
- Additional revenue: premium curated problem lists, enterprise connectors, consulting / implementation templates.
- Upsell points: team licenses, advanced AI packs (faster compute, dedicated models), historical export data.

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## 16. Edge case handling & reliability

- Duplicate detection tolerance and manual merge UI.
- Stale problem aging: problems older than X months flagged; re-evaluate Pi calculation.
- Data drift: retrain or reweight source authority monthly.
- Rate limits & backoffs for scrapers.
- Fallback UX when LLM fails: show template + manual edit option.
- Runbooks for sensitive domains (health/finance) with mandatory compliance warnings and “consult an expert” CTA.

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## 17. Acceptance criteria (how to know it's done — MVP)

- Field onboarding saves and affects discovery results.
- Discovery pipeline returns  $\geq 50$  unique problem clusters per domain in seeded test domains.

- Pain Index ranking correlates with manual human ranking (A/B test with 3 domain experts — target Spearman  $\rho > 0.6$ ).
  - Solution generation produces a layered output consistently (schema verification).
  - Roadmap creation button produces an exportable roadmap with  $\geq 4$  phases and estimated times.
  - PRD and Tech Doc generators produce Markdown that passes a linter for required sections and includes at least one API stub.
  - Auth + subscription flows functional; basic billing via Stripe in sandbox.
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## 18. KPIs & success metrics

- Activation: % users who create a roadmap in their first 7 days.
  - Time-to-first-roadmap: median time from signup to roadmap generation.
  - Document generation rate: docs generated per active user / month.
  - Validation conversion: % of roadmaps where the discovery experiments produce positive validation.
  - Retention: 7/30/90-day retention for Pro users.
  - Revenue: MRR, ARPU, churn.
  - Content quality: user rating of generated docs (1–5) and edit distance to final saved doc.
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## 19. Implementation plan (high level)

**Sprint 0 (2 weeks):** Infra, auth, data model, minimal discovery connectors (1–2 domains), LLM evaluation.

**Sprint 1 (4 weeks):** Discovery pipeline + Pain Index + problem feed UI.

**Sprint 2 (4 weeks):** Layered solution generation + solution view.

**Sprint 3 (3 weeks):** Roadmap engine + export.

**Sprint 4 (3 weeks):** PRD & Tech Doc generator, inline editor.

**Launch (beta):** invite-only with feedback loop and metrics.

Iterate post-beta: payments, team features, more connectors.

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## 20. Risks & mitigations

- **Low-quality scraped data** → strong dedupe, source authority map, manual curator UI.
- **LLM hallucination / incorrect technical suggestions** → include deterministic templates, require code/diagram stubs to be validated, flag low-confidence segments.
- **Regulatory risk in domains like healthcare** → require disclaimers, avoid clinical advice, push to expert review.



- **Cost of model usage** → caching, rate-limits, token budgets, selective use of smaller models for extraction.
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## 21. Future roadmap (beyond 1st year)

- Team workspaces, repo scaffolding & CI templates (auto-init GitHub repos).
  - Deep GitHub integration for PR templates and automated scaffolding.
  - Community marketplace for curated problem sets and solution templates.
  - Multi-modal ingestion (images, PDFs) and in-product experiment tracking.
  - On-prem or private model support for enterprise customers.
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## 22. Tone, quality, and deliverable standards

- Outputs must be concise, actionable, and non-fluffy. Use sectioned templates with checklists and explicit acceptance criteria.
  - All generated artifacts (PRD, Tech Doc, Roadmaps) must include at least one *measurable* success metric and next actionable step.
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## 23. Immediate next steps (for team)

1. Wireframe onboarding + problem feed (design spike).
  2. Implement one reliable discovery connector and the Pain Index pipeline (engineering spike).
  3. Select primary LLM provider and build model orchestration shim.
  4. Build Roadmap generator template & export pipeline.
  5. Prepare beta cohort & instrumentation.
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## 24. Deliverables to produce now

- Production PRD (this doc).
- Roadmap templates (phase templates + JSON schema).
- Document generator templates (PRD, Tech Doc, Auth Flow).
- Minimal API contract for export (CSV, OpenAPI stub).